

Economic Forecasting for Banking Using FRED Time Series: Phase 1 — Exogenous Macro Signal for the 30-Year Mortgage Rate

Final Report

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GitHub: https://github.com/lynchpo/dragon_exploder_fred

Abstract

Banks need a clear distinction between last week's mortgage rate and information that can be recovered from public macroeconomic series alone. This Phase 1 study asks whether a leakage-controlled set of exogenous FRED indicators can support chronological forecasts of the weekly 30-year fixed mortgage rate, and under what conditions that signal weakens. Mixed-frequency series are aligned to a weekly calendar; features are lags, rolling statistics, and regime indicators of exogenous series only. The winning LightGBM configuration (candidate C) was locked using training-window validation mean squared error, then scored once on a sealed 2021–2026 chronological hold-out, where it reaches test $R^2 \approx 0.85$ and outperforms regularized linear and SARIMAX baselines. A random walk remains the stronger benchmark for a pure one-week-ahead point forecast. Hold-out errors concentrate in 2022–2023. That pattern is consistent with a shift in the mortgage–Treasury relationship documented in policy research, but it does not by itself prove that mechanism or rule out omitted drivers. Monte Carlo scenario paths remain Phase 2.

1. Introduction

1.1 Background

The 30-year fixed mortgage rate is a primary channel from monetary policy into household borrowing costs, refinancing, and bank mortgage-asset valuation. McCormick and Ramaswamy (2026) show that this channel is not a simple map from the federal funds rate: in their decomposition, the 10-year Treasury yield, the slope of the curve, and implied rate volatility explain most variation in mortgage spreads over Treasuries, with a much larger partial response to the 10-year yield than to the funds rate. That fact is the business case for an exogenous-macro pipeline aimed at risk and treasury desks rather than at day-trading last week's print.

The 2022–2023 tightening cycle made the distinction urgent. Brookings analysis of the post-2020 increase documents that 30-year mortgage rates rose by more than long-term Treasuries over part of that episode (Gamber et al. 2024). Drechsler, Savov, Schnabl, and Supera (2024) attribute a large share of the fall and subsequent rise in the mortgage–Treasury spread to bank and Federal Reserve mortgage-backed securities supply during easing and tightening, not only to the 10-year yield itself. Justiniano, Primiceri, and Tambalotti (2017) show that mortgage–Treasury disconnects have occurred before. Together these sources motivate a bank-facing question that is not “can a model beat persistence at one week?” but “how much macro information remains when prior mortgage rates are withheld, and where does that information fail?”

1.2 Research question

Can a leakage-controlled set of exogenous FRED macro series support chronological forecasts of the weekly 30-year fixed mortgage rate for bank risk and treasury use, and under what conditions does that signal weaken?

1.3 Hypothesis and prediction

Hypothesis. Treasury yields and related policy macros carry usable predictive information for mortgage rates even when lagged mortgage rates are excluded. Forecast errors should concentrate when the mortgage–Treasury relationship is under stress, as in rapid tightening.

Prediction. An exogenous-only gradient-boosting model will outperform regularized linear and SARIMAX baselines on a post-2021 chronological hold-out. A random walk will remain competitive or superior for pure one-step persistence. Mean residuals will be positive in the 2022–2023 window. Those residuals are expected to be consistent with a relationship shift; they are not predicted to identify a single structural cause.

1.4 Two stakeholder questions

If the question is “What is the best one-week-ahead forecast of the mortgage rate?”, the random walk is the better benchmark. If the question is “How much can we forecast from macroeconomic information without using prior mortgage rates?”, LightGBM is the better exogenous comparator (Geist, baseline feedback). Phase 1 answers the second question and reports the first so a stakeholder is not turned around.

2. Data

2.1 Acquisition and suitability

Series come from the Federal Reserve Bank of St. Louis Federal Reserve Economic Data (FRED) repository (Federal Reserve Bank of St. Louis). Public graph CSVs are reproducible without a proprietary vendor. The target is MORTGAGE30US, the weekly average 30-year fixed mortgage contract rate. Core predictors follow the channels emphasized by McCormick and Ramaswamy (2026) and standard pass-through practice: DGS10 (10-year Treasury), FEDFUNDS (policy rate), UNRATE (labor slack), CPIAUCSL (price level), and HOUST (housing activity). Engineered fields include the yield spread (DGS10 – FEDFUNDS), lags and rolling statistics of exogenous series only, rates of change, and regime indicators. After alignment and feature construction the modeling matrix is on the order of 1,300 weekly rows and about 60 exogenous predictors. The codebook is Appendix A.

Collection caveat. FRED series are revised. The observation date on a current download is not automatically the historical release date a real-time forecaster would have seen. Full vintage reconstruction (ALFRED) is out of scope for this capstone. Conservative publication-lag rules are stated in Section 2.2.

2.2 Cleaning, alignment, and publication-lag rules

MORTGAGE30US is weekly; DGS10 is daily; FEDFUNDS, UNRATE, CPIAUCSL, and HOUST are monthly. All series are resampled to a weekly Friday grid. Daily Treasuries are reduced to the weekly stamp without using future calendar days. Monthly prints are forward-filled after they appear on that calendar. Forward-fill after a monthly value is known avoids look-ahead from later months; last-observation-carried-only-within-month and explicit multi-week publication lags were considered as alternatives. The working rule is: a monthly observation is treated as available from the first weekly

timestamp at or after month-end. That is conservative relative to ignoring lags and is not a claim of official BLS/Fed release-day realism. Missing values created by lag and rolling windows at the start of the sample are dropped. No MORTGAGE30US-derived predictors enter reported models.

2.3 Exploratory analysis on the final analytical matrix

Early EDA used raw prints. Following instructor guidance, exploration here refers to the frequency-aligned, lagged, and engineered matrix used for modeling. Contemporaneous raw correlations can shift after lagging and alignment; conclusions below are therefore drawn from the working feature set.

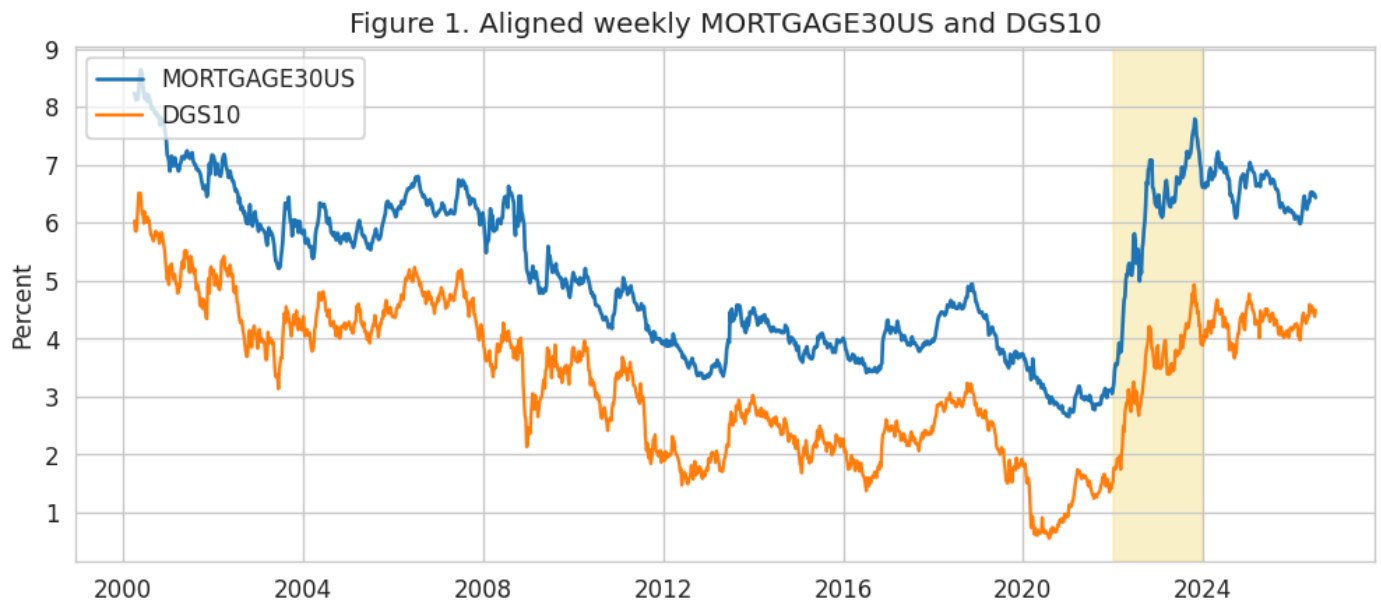


Figure 1 motivates both the Treasury-led hypothesis and the stress-window evaluation rule.

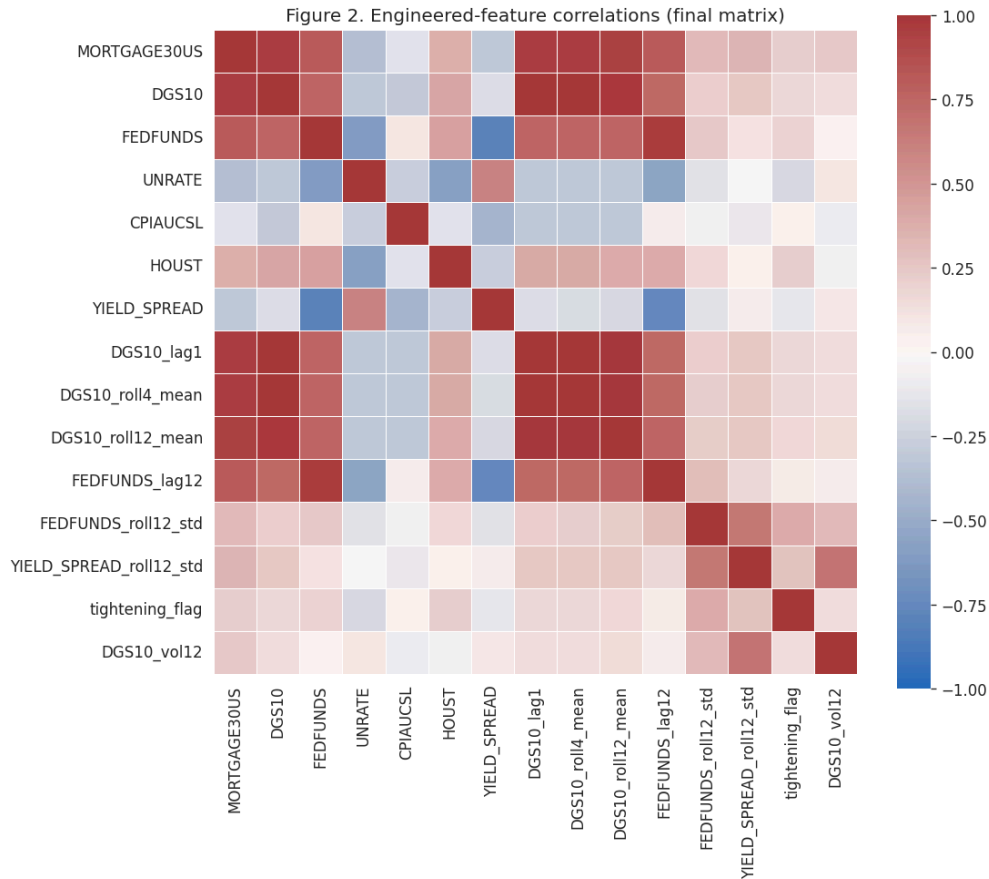


Figure 2 is consistent with McCormick and Ramaswamy (2026) on the primacy of the 10-year channel, without treating correlation as forecast skill.

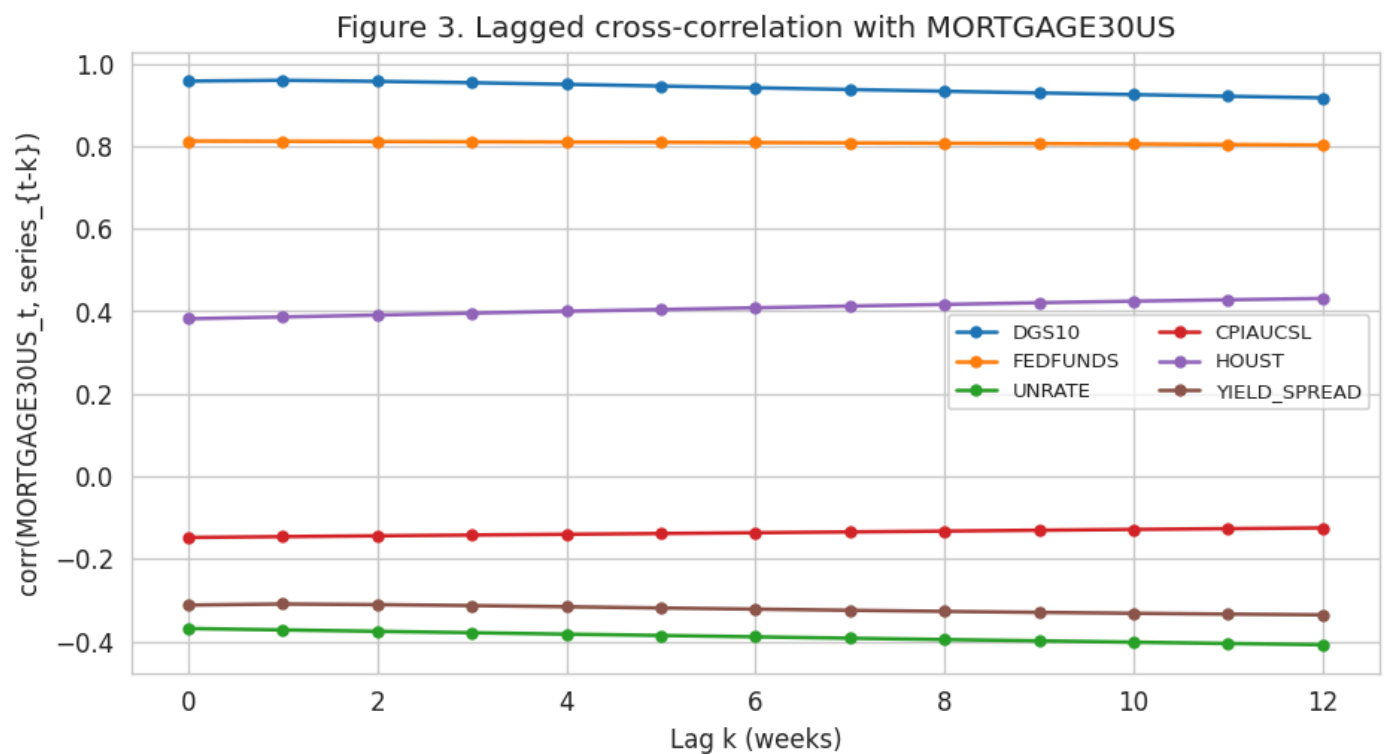


Figure 3 replaces contemporaneous-only EDA for a true forecasting design.

Figure 4. Mortgage-Treasury spread (aligned weekly)



Figure 4 is the descriptive counterpart of the 2022–2023 residual story and of the spread literature (Gamber et al. 2024; Drechsler et al. 2024).

Figure 5. Tightening flag over time

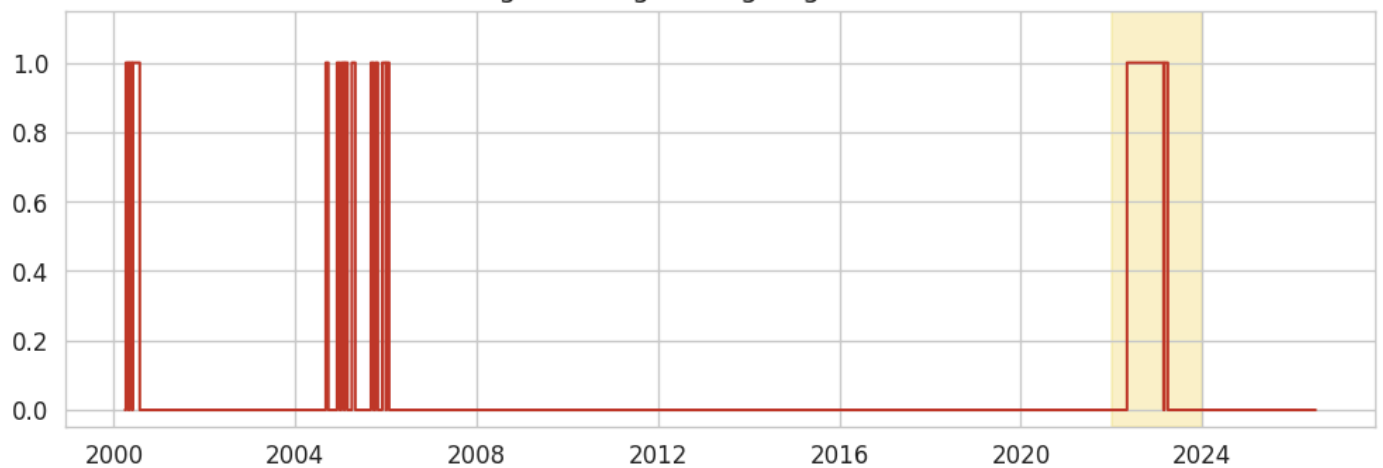


Figure 5 documents why a co-primary stress metric was added rather than relying only on full-sample R^2 .

3. Models

3.1 Preprocessing and feature engineering

Reported models share one exogenous matrix: weekly alignment and forward-fill; yield spread; lags at 1, 2, 4, 8, and 12 weeks; rolling means and standard deviations; rates of change; tightening flag, 12-week Treasury volatility, and 12-week funds-rate change. Target lags and target rolling statistics were built only for an ablation that documented leakage (hold-out R^2 jumped toward 0.98; correlation of a four-week target rolling mean with the same-week target ≈ 0.997) and were then excluded. Feature screening and block tests used to justify that exclusion and the lag design were conducted inside the training period. One-hot encoding was unnecessary.

3.2 Unsupervised exploration

A two-component Gaussian mixture model on policy-change, volatility, and curve features produced a soft stress probability as a complement to the hand flag. PCA on the exogenous matrix described variance concentration. Supervised models kept named macro features rather than principal components so bank readers could see Treasury versus housing channels.

3.3 Algorithms, assumptions, and baselines

Baselines are a random walk (last week's rate), OLS, SARIMAX(1,1,1) with a reduced exogenous set, Bayesian Ridge, and LassoCV. The primary non-linear model is LightGBM (Ke et al. 2017). Linear residual ACF, Ljung–Box tests, and Durbin–Watson statistics show highly persistent level bias rather than a need for a richer autoregressive error as the main fix. LightGBM does not use those linear residual assumptions; it also cannot extrapolate beyond training ranges if rates enter unprecedented territory such as an explicit limit under regime shift.

3.4 Lock rule, tuning, and validation

Lock rule. Hyperparameters and the winning configuration were selected using only the training window and a train-internal validation slice. Candidate C was locked before the 2021–2026 chronological test was used for reporting. That hold-out is evaluated once for confirmation. Tables that compare A versus B versus C, or restricted versus full feature sets, on the test window are descriptive after lock and were not the rule that chose the winner.

Two search strategies used the same grid: learning rate in {0.03, 0.05, 0.08}, max depth in {3, 4, 5}, num leaves in {15, 31}. Strategy B used early stopping on the last 20 percent of train (a relatively calm slice). Validation R^2 on that slice was poor because target variance was low; early stopping truncated capacity and was rejected on train-internal grounds. Strategy C fixed 500 trees and ranked by validation MSE instead of validation R^2 . The train-only winner was learning rate 0.05, max depth 4, 31 leaves, 500 trees, subsample 0.8, column sample 0.8, random_state 42 (candidate C / C_fixed500), trained on the full exogenous matrix.

Candidate	Test R^2	Test MSE	Mean resid	Stress resid	Stress MSE	Score
C fixed-500	0.845	0.320	0.400	0.714	0.587	0.906
A pre-tune	0.833	0.343	0.417	0.727	0.617	0.960
B early-stop	0.789	0.435	0.506	0.823	0.769	1.203

Table 1. Descriptive confirmation metrics on the sealed 2021–2026 hold-out after C was locked from train-internal validation MSE. Score = Test MSE + 2022–23 stress MSE. This table does not elect the winner.

On the same locked hyperparameters, a descriptive feature ladder on the hold-out gave test $R^2 \approx 0.73$ for DGS10_lag1 only, ≈ 0.79 for a Treasury core, and ≈ 0.85 for the full exogenous set. Extra macros improve overall fit more than they close the 2022–2023 residual gap.

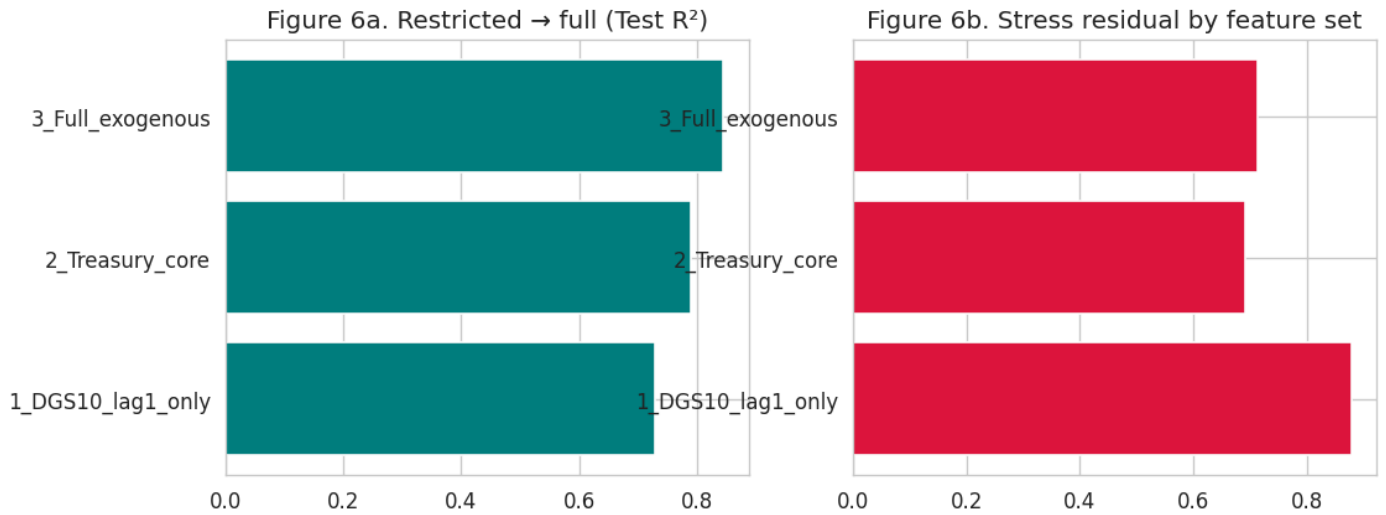


Figure 6. Restricted-to-full feature ladder on the sealed 2021–2026 hold-out using locked LightGBM C hyperparameters. Left: test R^2 rises from a single 10-year lag through a Treasury-core block to the full exogenous set. Right: 2022–2023 mean residual stays large even as the feature set expands. Extra macros improve average-week fit more than they close the stress-window gap. This comparison is descriptive; it was not used to choose C .

Figure 7. Sealed chronological hold-out: actual weekly 30-year mortgage rate versus locked LightGBM C_{fixed500} . The gold band is 2022–2023. The model tracks the early test period, then systematically under-predicts through tightening (residual mean ≈ 0.40 overall). Residuals shrink again after 2024. This path is the locked confirmation plot, not an early-stopping run.

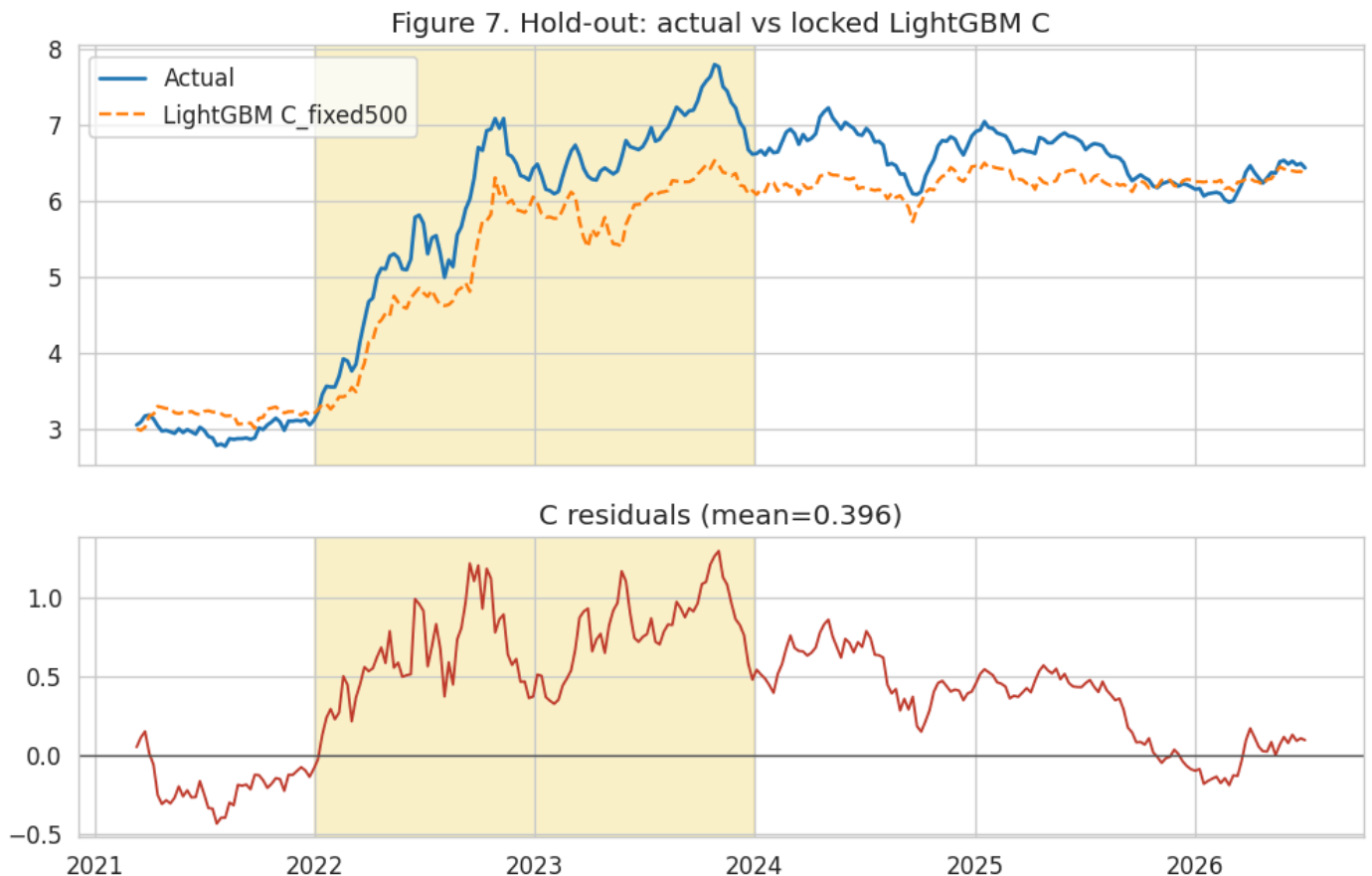


Figure 8. SHAP drivers by regime (C_fixed500)

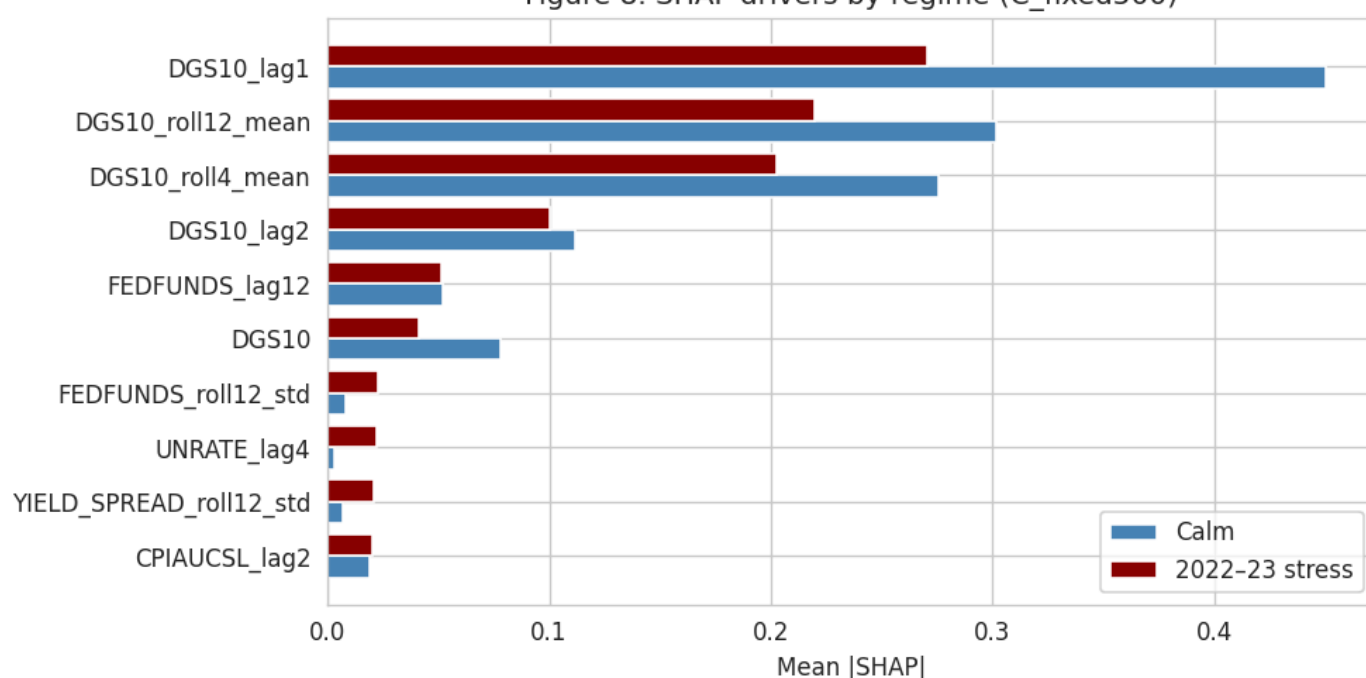


Figure 8. Mean absolute SHAP values for locked C on the same hold-out, split into calm weeks versus 2022–2023. Lagged and rolling 10-year Treasury features rank first in both regimes. Several Treasury SHAP magnitudes are smaller under stress, which is consistent with a weaker mortgage–Treasury mapping in that window rather than a switch to a different driver set.

3.5 Interpretation of the locked model

Split importance and SHAP rank DGS10_lag1 and rolling Treasury means first, then longer Fed lags and secondary macros—aligned with Lasso’s sparse pattern and with McCormick and Ramaswamy (2026). SHAP magnitudes on leading Treasury features are smaller in stress weeks than in calm weeks, yet the ranking stays Treasury-led. The model still under-predicts in 2022–2023 (mean residual $\approx +0.40$ overall; $\approx +0.71$ in the stress window).

That residual pattern is consistent with a mortgage–Treasury relationship shift of the kind described by Gamber et al. (2024) and with supply-side spread moves in Drechsler et al. (2024). Justiniano et al. (2017) show such disconnects can persist historically. These results do not, by themselves, prove a single disconnect mechanism or rule out omitted drivers such as MBS demand, quantitative tightening, origination capacity, or credit-risk premia.

Across families: the random walk is the best pure one-step persistence benchmark; Lasso is the best linear exogenous model from the baseline week; LightGBM C is the best exogenous non-linear model under the train-only lock. Hold-out comparisons across families are descriptive once each family is locked.

3.6 Stakeholder Results

For a risk or treasury desk, the useful distinction is not “did the model fit last week’s rate?” but “if we are not allowed to look at last week’s mortgage print, which public macros still move the forecast, and when does that map break?”

Split importance and SHAP give the same first-order answer. The one-week lag of the 10-year Treasury yield (DGS10_lag1) and short rolling means of that yield dominate. Longer federal-funds lags and secondary series; inflation, unemployment, housing starts, enter later and with smaller contributions. That ordering matches the sparse Lasso coefficients from the baseline week and the pass-through evidence in McCormick and Ramaswamy (2026): mortgage rates track the 10-year much more than they track the policy rate one-for-one. The model is listening to the bond market's long rate, not inventing a housing-only story.

Figure 8 splits those SHAP magnitudes into calm weeks versus 2022–2023. The *ranking* does not flip: Treasuries remain first in both regimes. The *size* of the Treasury contributions shrinks in the stress window. The model is still using the same channel; that channel simply explains less of the mortgage print when the market is tightening fast. Figure 6 makes the same point with features instead of SHAP: adding the full exogenous set raises overall hold-out R^2 relative to a Treasury-only block, but it barely reduces the 2022–2023 residual. Extra macros help the average week. They do not buy you the spike.

What the desk actually sees on the hold-out (Figure 7) is a mean residual of about +0.40 percentage points overall and about +0.71 points in 2022–2023. Positive residual means the locked model came in *low*. Through the gold band the actual contract rate ran ahead of what contemporaneous Treasuries and the rest of the FRED block could justify under a relationship estimated mostly on earlier regimes. After 2024 the gap narrows again. That is a break-detector pattern, not a claim that C is a bad model in every week.

That pattern is consistent with a change in the mortgage–Treasury *spread*, not only in the 10-year level. Gamber et al. (2024) document an unusually large 2022–early 2023 widening of 30-year mortgage rates relative to long Treasuries. Drechsler et al. (2024) show that bank and Federal Reserve MBS supply compressed the spread in the 2020–21 easing and widened it again when policy reversed. Justiniano, Primiceri, and Tambalotti (2017) show that mortgage rates have disconnected from Treasuries before. Those papers supply the field context for *why* a Treasury-led model can look well-behaved and still miss a tightening cycle. They do not let this notebook name a single cause. MBS demand, quantitative tightening, origination capacity, servicing and convexity costs, and credit-risk premia are all omitted from the FRED feature list. Phase 1 can say the residual is consistent with a relationship shift. It cannot adjudicate among those channels.

How to use the family of models without mixing up the questions:

- If the question is “what is the best one-week-ahead *number*?”, use the random walk. Last week's rate is still the persistence floor.
- If the question is “how much can we say from macros alone, with no lagged mortgage rate?”, Lasso is the honest linear answer from the baseline week, and locked LightGBM C is the stronger

non-linear exogenous answer (hold-out $R^2 \approx 0.85$, versus Lasso ≈ 0.42 on that same leakage-controlled design).

- If the question is “is the historical pass-through still working this quarter?”, watch Figure 7’s residual and Figure 4’s spread together. A *rising spread* plus a *large positive C residual* is the operational flag that the Treasury map is stale.

Hold-out tables that also show pre-tune or early-stop LightGBM, or that compare families after each was locked, are confirmation, not a second bake-off. C was selected on train-internal validation MSE. The stakeholder takeaway is therefore limited and usable: public Treasuries carry real planning signal when leakage is blocked; that signal is not a substitute for last week’s print at a one-week horizon; and 2022–2023 is the documented week-set where a desk should not treat the historical mapping as intact.

4. Project-Specific Guidance Incorporated

Instructor notes from EDA through tuning are answered in this document as follows.

- Sealed hold-out: C locked on train-internal validation MSE; 2021–2026 scored once; A/B/C and ladder tables labeled descriptive (PFE and tuning notes).
- Publication lag: revised FRED $\neq$ release date; monthly series treated as available from the first weekly stamp at or after month-end; ALFRED out of scope.
- Forward-fill after a known monthly print, with alternatives considered.
- EDA on the aligned and engineered matrix (Figures 1–5), including lagged cross-correlation (Figure 3).
- Regime indicator features after EDA; used in modeling and stress metrics.
- Stakeholder split: random walk for one-week-ahead; LightGBM for exogenous-only signal.
- 2022–23 residuals framed as consistent with a relationship shift, not as proof; omitted drivers acknowledged.
- Phased delivery: Monte Carlo remains Phase 2.

5. Discussion and Next Steps

5.1 Conclusions

Relative to the research question, exogenous FRED macros support a usable chronological signal when leakage is controlled. The hypothesis that Treasury and policy information matter is supported by the train-locked LightGBM confirmation metrics, the descriptive feature ladder, and SHAP rankings that follow the 10-year channel in McCormick and Ramaswamy (2026). The hypothesis that errors concentrate under rapid tightening is supported by stress-window residuals. The prediction that boosting beats linear exogenous baselines holds. The prediction that a random walk remains the pure one-step champion also

holds. Persistent 2022–2023 under-prediction is consistent with the spread episode in Gamber et al. (2024) and Drechsler et al. (2024) and with earlier disconnect evidence in Justiniano et al. (2017); it is not a standalone identification of cause.

5.2 Contribution

The contribution is a reproducible Phase 1 pipeline that separates persistence from exogenous macro skill, documents target-lag leakage, and reports regime breakdown in language a treasury desk can use. Against classical SARIMAX and linear exogenous baselines on this hold-out, locked LightGBM (Ke et al. 2017) improves exogenous fit while remaining inspectable through SHAP and restricted feature sets.

5.3 Recommendations

Use Phase 1 as a map, not as a one-week price. Treasury and risk teams should keep three outputs on the same dashboard: last week's mortgage rate (persistence), locked LightGBM C (exogenous macro signal), and the mortgage–Treasury spread plus C's residual (break detector). Act on disagreement among those three, not on C alone. Do not replace a random-walk short-horizon mark with the booster.

Treat 2022–2023 as the design constraint for Phase 2. The residual mean of about +0.4 overall and about +0.7 in the tightening window is the requirement Phase 2 has to beat or explain. Adding more FRED columns is unlikely to be the main fix: Figure 6 already showed that a wider exogenous set lifts average-week R^2 more than it closes the stress gap. Phase 2 should change the question and the output, not only the feature list.

Phase 2 experiments:

Monte Carlo paths on the residual and on the spread, not on a single next-week point. Simulate many 4-, 12-, and 26-week paths whose shocks are drawn from hold-out residuals, with a stress-weighted draw that oversamples 2022–2023-like weeks.

Insight available: a distribution of “how high could the contract rate get if pass-through stays broken,” and a probability that the rate exceeds a desk threshold (for example 7%). Phase 1 cannot say that; it only scores one realized path.

Regime-conditional or two-step paths (Treasury level + spread). First path the 10-year; second path the mortgage–Treasury spread using volatility, funds-rate change, and a stress flag. Add the two.

Insight available: whether the 2022–23 miss was a level failure or a spread failure. If most of the +0.7 lives in the spread step, MBS supply and origination-capacity data become the next build; if it lives in the Treasury step, the problem is extrapolation of DGS10, not housing macros.

Multi-step evaluation (4- and 12-week direct models). Persistence will weaken as the horizon lengthens; that is where exogenous macros can earn their keep.

Insight available: a horizon at which C (or a direct 12-week model) is no longer dominated by the random walk. That number is what a planning calendar actually needs. Phase 1's one-week walk-forward was never that test.

Real-time vintages (ALFRED) and explicit publication lags. Rebuild the same C pipeline on first-release prints.

Insight available: how much of the Phase 1 $R^2 \approx 0.85$ is revised-data luck. If vintage R^2 falls sharply, the dashboard must carry a lag disclaimer; if it holds, the Phase 1 map is closer to operational.

Interval calibration under stress. Residual bootstrap or quantile regression, scored separately on calm weeks and on 2022–23. Phase 1 bands were too narrow in the spike.

Insight available: a coverage number a risk committee can audit (“90% of stress weeks fell inside the band” or they did not). Without that, Monte Carlo widths are decoration.

Decision rule for “Phase 2 worked.”

A successful next phase produces (a) multi-horizon error versus a random walk, (b) stress-window residual materially below +0.7 or an honest interval that contains the miss, and (c) at least one documented mechanism test (spread step versus omitted MBS supply). Until those three exist, the recommendation from Phase 1 stands: use C as a Treasury-led signal and a stale-mapping alarm, and keep last week's rate as the one-week number.

5.4 Limitations

Revised FRED prints are not real-time information sets; publication-lag error may inflate apparent skill. Stress weeks are relatively rare, so performance is era-dependent. Tree models do not extrapolate beyond training support. Residual autocorrelation means point forecasts omit persistent structure. One-step random-walk dominance means exogenous R^2 must not be marketed as short-horizon trading accuracy. Weekly aggregation, forward-fill availability, and hand stress flags are simplifying assumptions; if they fail in production, operational reliability changes. Phase 1 is signal recovery and risk narrative support, not an autonomous pricing engine.

6. Code Availability

Code, locked configuration, and reports: https://github.com/lynchpo/dragon_exploder_fred. Notebooks are under notebooks/. Locked helpers are under src/ (phase1_locked_model.py, phase1_feature_list.txt, phase1_locked_config.json, MODEL_CARD_PHASE1.md). Dependencies are in requirements.txt. LightGBM random_state is 42. FRED public series are downloaded in-notebook.

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Appendix, Data Dictionary

Variable / family	Type	Description
MORTGAGE30US	Target	Weekly 30-year fixed mortgage rate (FRED)
DGS10, FEDFUNDS, UNRATE, CPIAUCSL, HOUST	Exogenous levels	Treasury, policy, labor, inflation, housing starts
YIELD_SPREAD	Engineered	DGS10 minus FEDFUNDS
*_lag1,2,4,8,12	Engineered	Lags of exogenous series only
_roll / *_roc*	Engineered	Rolling statistics and rates of change (exogenous)
tightening_flag, DGS10_vol12, FEDFUNDS_change_12w	Regime	Stress indicators and policy impulse

Table A1. Modeling-matrix codebook. No MORTGAGE30US_* predictors are used in reported models. Locked configuration C_fixed500: LightGBM learning_rate=0.05, max_depth=4, num_leaves=31, n_estimators=500, subsample=0.8, colsample_bytree=0.8, random_state=42.